
Seminar: Fermat and Number Theory

Winter 2026/27

Content

This seminar offers an elementary introduction to modern number theory, starting with the fascinating work of Pierre de Fermat in the 17th century. Questions raised by Fermat have been occupying mathematicians for centuries, culminating in Wiles' proof of the so-called "Fermat's last theorem". Efforts to resolve these questions have led mathematicians to discover and explore vast new areas in the wonderland of numbers.

We will start the seminar by studying Fermat's argument for the fact that there are no positive integers a , b , and c , satisfying $a^4 + b^4 = c^4$. His solution implicitly uses elliptic curves, providing us with a natural motivation to introduce these fundamental objects. We then move to p -adic numbers which are an essential tool in much of the recent progress in number theory, and use them to study rational points on conics. Next, we study one of the most important and mysterious functions in all of mathematics: the Riemann zeta function. Finally, we give an elementary introduction to algebraic number theory and conclude the seminar with Kummer's celebrated work on Fermat's last theorem.

Preliminary Meeting: 21.07.26, 2:15 pm, SR 4

Seminar data:

Target audience:	Bachelor students	Schedule:	Tuesdays 14-16
Lecturers:	Judith Ludwig Alireza Shavali Miriam Philipp	Start:	13.10.26
		Place:	TBA
		Code:	MSB
Language:	English	Credit:	6 LP

References:

- Kato K., Kurokawa N., Saitō T., *Number Theory 1: Fermat's dream*. Translations of Mathematical Monographs, AMS, 2000.